Test Series No.

01

All India Test Series
NEET (2025-26)

DURATION: 180 Minutes | **DROPPER** | **M. MARKS: 720**

DATE: 14/09/2025

SYLLABUS

Physics:	Units and Measurements, Mathematical Tools, Motion in a Straight Line, Motion in a Plane
Chemistry:	Some Basic Concepts of Chemistry, Redox Reaction
Botany:	Cell: the unit of life
Zoology:	Structural Organization in Animals (including Frog & cockroach)

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

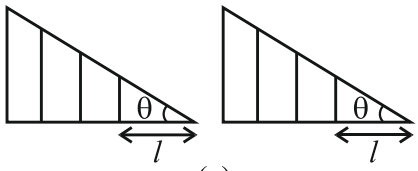
Name of the Student (In CAPITALS): _____

Roll Number: _____

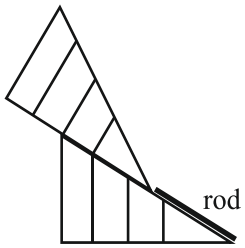
OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. A student designs an improvised vernier caliper using two identical wedge shaped scales as shown in figure (a), reading of length of a rod is taken as shown in figure (b). Assume perfect alignment and no zero error, what is the least count of this improvised vernier caliper? (each divisions are equally spaced = l)



(a)



(b)

(1) l (2) $l \tan\theta$

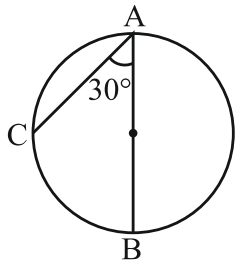
(3) $l \sec\theta - l$ (4) $l \cos\theta - l$

2. For a particle moving in a straight line, velocity v as a function of position x is given by $v(x) = 4 - 2x$. Given that at $t = 0$, $x = 0$ which of the following equation, gives position x of particle as function of time t?

(1) $x = 2e^{-2t}$ (2) $x = 4e^{-2t}$

(3) $x = 2(1-e^{-2t})$ (4) $x = 4(1-e^{-2t})$

3. Two wires AB (diameter) and AC are fixed on a vertical circular loop of radius R. A small frictionless bead is released from rest along the two wires one by one from A. The **incorrect** statement(s) is/are: (g is acceleration due to gravity)



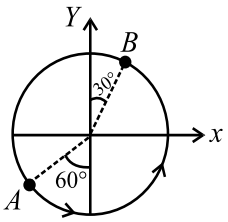
(1) Time taken to reach the points B and C from A is same

(2) Time taken to reach the point C is $\sqrt{\frac{4R}{g}}$

(3) Speed of bead at B and C are same

(4) Both (2) and (3)

4. A particle is moving in a circular path of radius R with constant speed v. During the time interval, when it moves from A to B as shown in figure, the magnitude of average acceleration is equal to:



(1) $\frac{\left(\sqrt{3}+1\right)v^2}{7\sqrt{2}\pi R}$

(2) $\frac{6v^2}{7\sqrt{2}\pi R}$

(3) $\frac{6\left(\sqrt{3}+1\right)v^2}{7\sqrt{2}R}$

(4) $\frac{6\left(\sqrt{3}+1\right)v^2}{7\sqrt{2}\pi R}$

5. A particle is moving in x-y plane on path $y = 2x - x^2 + 4$, where x and y are in metre. Its x component of velocity is constant and is equal to 4 m s^{-1}. Acceleration of the particle at $x = 1$ m is

(1) -4 m s^{-2} (2) -8 m s^{-2}

(3) -32 m s^{-2} (4) -16 m s^{-2}

6. In an experiment to determine acceleration due to gravity g, a student measures length of simple pendulum $l = (100.0 \pm 0.1)$ cm. He records time for 20 oscillations as (40 ± 1) s. Then, **correct** statement is:

(1) Error ΔT in time period T is 1 s.

(2) Maximum percentage error in time period T is 5%

(3) Maximum percentage error in the measurement of g is 2.6%

(4) Maximum percentage error in the measurement of g is 5.1%

7. A particle is projected in air with speed u at an angle θ_0 with horizontal. The magnitude of change in magnitude of velocity between the point of projection and the maximum height is:

(1) u

(2) $u \sin\theta_0$

(3) $u(1 - \sin\theta_0)$

(4) $u(1 - \cos\theta_0)$

[Contd...

8. A physical quantity P is dependent on other physical quantities a, b, c and d respectively as $P = \frac{a^2 b}{\sqrt{cd^3}}$.

If percentage errors in a, b, c and d are 2%, 1%, 1% and 2% respectively then maximum percentage error in P will be:

(1) 8.5%

(2) 11.5%

(3) 6.5%

(4) 7%

9. The value of $\frac{d}{dx}$ [cos (60°)] is:

(1) $\frac{\sqrt{3}}{2}$

(2) $\frac{1}{2}$

(3) $\frac{1}{\sqrt{2}}$

(4) 0

10. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): In projectile motion, magnitude of rate of change of velocity is variable.

Reason (R): In projectile motion, speed first decreases then increases.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

11. A lift starts ascending from rest with constant acceleration of 2 m s^{-2}, then with constant velocity and finally comes to rest with constant retardation of 2 m s^{-2}. If total time taken is 10 s and total distance covered is 32 m, then maximum speed with which lift travels, is:

(1) 4 m s^{-1}

(2) 8 m s^{-1}

(3) 12 m s^{-1}

(4) 6 m s^{-1}

12. For a particle moving in a circular path with constant speed, at a given instant, its velocity is $\vec{v}$. Then correct statement is:

(1) $\left|\frac{d\vec{v}}{dt}\right| = \frac{d}{dt}|\vec{v}| \neq 0$

(2) $\frac{d|\vec{v}|}{dt} = 0; \left|\frac{d\vec{v}}{dt}\right| \neq 0$

(3) $\left|\frac{d\vec{v}}{dt}\right| = 0; \frac{d}{dt}|\vec{v}| \neq 0$

(4) $\left|\frac{d\vec{v}}{dt}\right| = \frac{d}{dt}|\vec{v}| = 0$

13. A screw gauge has least count 0.01 mm and the number of circular scale divisions is 100. The pitch of the screw gauge is

(1) 1 cm (2) 0.001 cm

(3) 0.1 cm (4) 0.0001 cm

14. A projectile is projected with same speed 20 m s^{-1} at two different angles of projections with horizontal such that their horizontal range is same in two cases. If H_1 and H_2 be the maximum heights respectively in two cases, then $(H_1 + H_2)$ is equal to (g = 10 m s^{-2})

(1) 40 m (2) 80 m

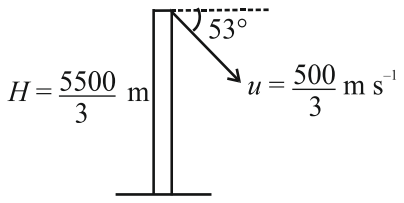
(3) 60 m (4) 20 m

15. Kinetic energy of a particle moving along x-axis as a function of its position (r) and time (t) is given as $K = \frac{Ar^{2/3}}{B + t^3}$. If force acting on the particle depends on A, B and length L as $F = A^x B^y L^z$, then the value of $x^2 - y + z$ is:

(1) $\frac{2}{3}$ (2) $\frac{5}{3}$

(3) $\frac{1}{3}$ (4) $\frac{4}{3}$

16. A particle is projected from the top of a tower as shown in figure, then the distance from the foot of the tower where it will strike the ground will be: (g = 10 m s^{-2})



(1) $\frac{4000}{3}$ m (2) $\frac{5000}{3}$

(3) 1000 m (4) 2000 m

[Contd...

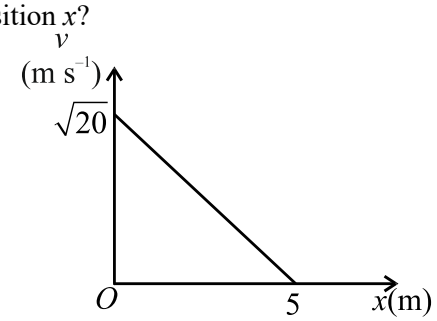
17. A particle is moving along the x axis. Its position (x) with time (t) is given by $x = 3t^2 - \frac{t^3}{2}$, where x is in metre and t is in second. Match the condition of list-I with time interval and instant of list-II.

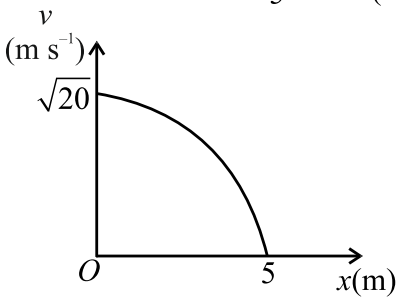
List-I		List-II	
A.	Velocity and acceleration will be in the same direction	P.	At $t = 0$ and $t = 4$ s
B.	Particle is at origin	Q.	$0 < t < 2$s
C.	Particle is retarding	R.	At $t = 0$ and $t = 6$ s
D.	speed is zero	S.	2 s $< t <$ 4 s

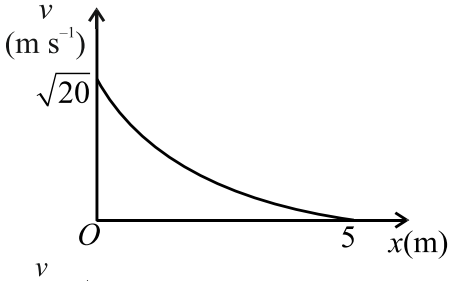
Choose the **correct** answer from the options given below.

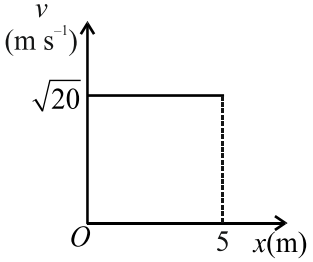
(1) (A)→(S), (B)→(P), (C)→(Q), (D)→(R)
(2) (A)→(Q), (B)→(P), (C)→(S), (D)→(R)
(3) (A)→(R), (B)→(S), (C)→(P), (D)→(Q)
(4) (A)→(Q), (B)→(R), (C)→(S), (D)→(P)

18. A particle is moving in a straight line with velocity $v_0 = +\sqrt{20}$ m s^{-1} at $x = 0$. It is having a constant retardation of 2 m s^{-2}. Which of the following graph correctly represents variation of velocity v with position x?

(1)


(2)


(3)


(4)


19. A particle is thrown vertically upward with a velocity 40 ms^{-1} where air resistance produces a constant retardation of 2 m s^{-2}. It reaches the ground after some time, then (g = 10 m s^{-2})

(1) time of ascent, $t_a = 5\sqrt{\frac{2}{3}}$ s

(2) time of descent, $t_d = \frac{10}{3}$ s

(3) Maximum height reached, $H = \frac{100}{3}$ m

(4) Speed just before striking the ground is $v = 40\sqrt{\frac{2}{3}}$ m s^{-1}

20. Two vectors $\vec{A} = \alpha\hat{i} - 2\hat{j} + 4\hat{k}$ and $\vec{B} = \alpha\hat{i} - \alpha\hat{j} - 2\hat{k}$ are perpendicular to each other. The value (s) of α is/are:

(1) 2 (2) –4
(3) –2 (4) Both (1) and (2)

21. Diagonals of a parallelogram are represented by $\vec{D}_1 = (6\hat{i} + 4\hat{j})$ m and $\vec{D}_2 = (-4\hat{j})$ m respectively. Area of this parallelogram is:

(1) 24 m^2 (2) 12 m^2
(3) 16 m^2 (4) 18 m^2

22. In ground to ground projectile motion, the speed of a projectile at its maximum height is $\sqrt{\frac{6}{7}}$ times its speed at half the maximum height. The angle of projection of the projectile with horizontal is:

(1) 60° (2) 53°
(3) 30° (4) 45°

23. If a particle is moving with constant velocity then which of the following statements are **correct**?

(A) Instantaneous velocity is equal to average velocity in any time interval.
(B) Instantaneous speed is equal to magnitude of instantaneous velocity.
(C) Acceleration is zero
(D) Distance covered is equal to magnitude of displacement in any time interval

(1) (A), (B) and (D) only
(2) (A), (B) and (C) only
(3) (A), (B), (C) and (D)
(4) (A) and (B) only

[Contd...

24. A train moving with constant acceleration crosses an electric pole. When engine of the train crosses the pole, speed of train is 5 m s^{-1} and 10 m s^{-1} when end of the train crosses it. What is the speed of the train when two-third of the train has yet to cross the pole?

(1) $\sqrt{\frac{125}{2}}$ m s^{-1}

(2) $\frac{\sqrt{125}}{2}$ m s^{-1}

(3) $\frac{15}{\sqrt{3}}$ m s^{-1}

(4) $5\sqrt{2}$ m s^{-1}

25. Given below are two statements:

Statement I: The sum of numbers 436.32, 227.2 and 0.301 with appropriate significant figures is 663.8

Statement II: If total displacement of a particle is measured as (12.0 ± 0.1) m in time interval (3.0±0.1) s, then average velocity upto correct significant figures is (3.00 ± 0.10) m s^{-1}

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

26. If A, B and C are physical quantities having different dimensions, then which of the following expression may give a meaningful quantity?

(1) $\frac{A-B}{C}$

(2) $\frac{A+C}{B}$

(3) $AB + C$

(4) $\frac{B+C}{A}$

27. In an imaginary system, unit of force is 100 N, unit of length is 10 m and unit of time is 60 s. The unit of mass in the new system is:

(1) 3600 kg

(2) 36000 kg

(3) 360 kg

(4) 36 kg

28. The minimum value of function $y = x^3 - 12x$ is:

(1) –2

(2) 2

(3) –16

(4) 16

29. When a man moves on a horizontal straight road with velocity 3 m s^{-1}, rain appears to be falling vertically downword. When he increases his velocity to 6 m s^{-1}, rain strikes him at angle 30° with the vertical. The actual speed of rain w.r.t. ground is:

(1) $3\sqrt{3}$ m s^{-1}

(2) 6 m s^{-1}

(3) $6\sqrt{3}$ m s^{-1}

(4) 9 m s^{-1}

30. Evaluate $\int_0^{\pi/4} (\cos^2 x - \sin^2 x)dx$

(1) 0

(2) $\frac{1}{2}$

(3) $-\frac{1}{2}$

(4) 1

31. Given $y = 4x - 2x^2$, then slope of tangent to the curve at $x = 0$ and at $x = 1$ respectively, are

(1) 0, 4

(2) 0, 0

(3) 4, 0

(4) 4, 4

[Contd...

32. A *super* swimmer who can swim with speed 5 m s^{-1} in still water, wants to cross a 500 m wide river flowing with speed 4 m s^{-1}. If swimmer swims making an angle 127° with downstream, then drift of the swimmer when it reaches the opposite bank, is:

(1) zero

(2) 100 m

(3) 150 m

(4) 125 m

33. Given three vectors $\vec{A},\vec{B}$ and $\vec{R}$ such that $\vec{R}=\vec{A}+\vec{B}$. $\vec{R}$ is perpendicular to $\vec{A}$ and $|\vec{R}|=6$. If $|\vec{A}|+|\vec{B}|=18$; then the angle between $\vec{A}$ and $\vec{B}$ is:

(1) 135°

(2) 150°

(3) 127°

(4) 143°

34. A stone is thrown vertically upward with speed 20 m s^{-1} from the top of a tower of height 25 m. The average speed of the stone for its entire journey until it hits the ground, is (g = 10 m s^{-2})

(1) 6.25 m s^{-1}

(2) 5 m s^{-1}

(3) 20 m s^{-1}

(4) 13 m s^{-1}

35. Two cars A and B are moving on a straight road in the same direction with speeds 20 m s^{-1} and 10 m s^{-1} respectively, with B ahead of A by 5 m. Sensing a collision, driver of car B starts increasing the speed at the rate of 4 m s^{-2}. At the same instant what effort should driver of car A make to avoid a collision?

(1) He should continue with same speed.

(2) He should apply brakes to produce a minimum retardation of 10 m s^{-2}.

(3) He should apply brakes to produce a minimum retardation of 6 m s^{-2}.

(4) He should apply brakes to produce a minimum retardation of 20 m s^{-2}.

36. A particle is thrown with the speed u at an angle α with the horizontal. When the particle makes an angle β with the horizontal, its speed will be:

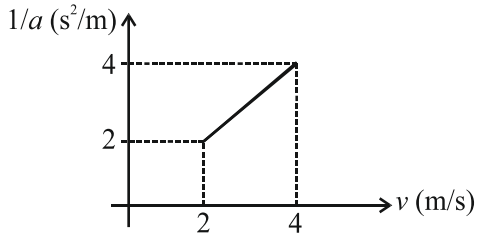
(1) u cosα

(2) u cosα secβ

(3) u cosα cosβ

(4) u secα cosβ

37. An ambulance moving along a straight line is rushing a patient to the hospital. The acceleration (a) of the ambulance depends on its speed (v). The graph below shows 1/a (in s^2/m) on the y-axis and speed v (in m/s) on the x-axis. Find the total time taken by the ambulance to increase its speed from 2 m/s to 4 m/s.



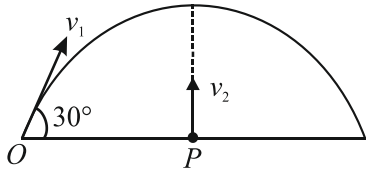
(1) 4 s

(2) 6 s

(3) 8 s

(4) 12 s

38. A particle is projected from point O with velocity v_1 at an angle 30° with the horizontal and at the same instant another particle is projected from point P which is vertically below the highest point with velocity v_2 in vertically upward direction as shown. For the two particles to collide the ratio $\frac{v_1}{v_2}$ will be:



(1) 1

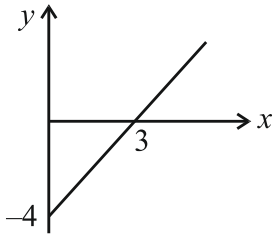
(2) 2

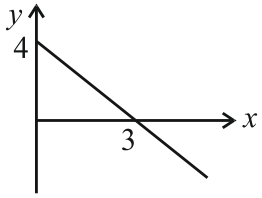
(3) 0.5

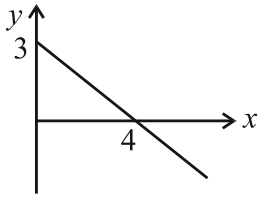
(4) 1.5

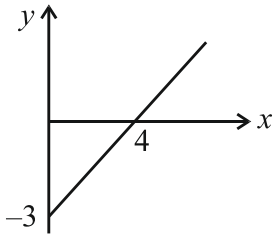
[Contd...

39. Which of the following graphs represents equation $\frac{x}{3}+\frac{y}{4}=1$?

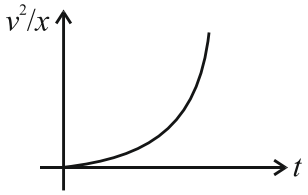
(1) 

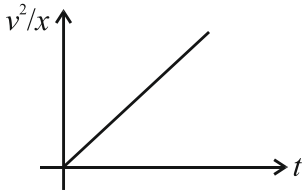
(2) 

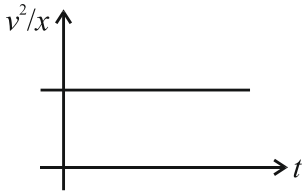
(3) 

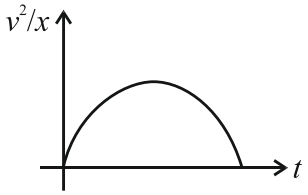
(4) 

40. A body starts from rest from origin and moves with constant acceleration. Let v be its velocity at time t and x be its displacement from origin at time t. Choose the **correct** graph between $\frac{v^2}{x}$ and t.

(1) 

(2) 

(3) 

(4) 

41. Acceleration a (in m s^{-2}) of an object falling through a fluid is given by $a = (8 - 2v)$ where v is speed of object in m s^{-1}. After a long time, it falls with constant speed. The constant speed is:

(1) 8 m s^{-1} (2) 4 m s^{-1}

(3) 2 m s^{-1} (4) 6 m s^{-1}

42. A boy wants to catch a bus at a distance 25 m away from him. Bus starts from rest with constant acceleration 2 m s^{-2}. The minimum constant speed with which boy should run to cath the bus is:

(1) 25 m s^{-1}

(2) 12.5 m s^{-1}

(3) 8 m s^{-1}

(4) 10 m s^{-1}

43. The significant figures in a measurement consist of:

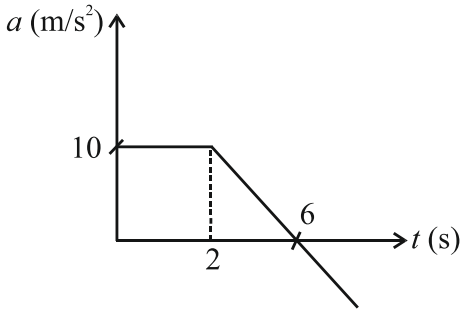
(1) All reliable digits only

(2) All reliable digits plus first uncertain digit

(3) All uncertain digits only

(4) First reliable digit and all uncertain digits

44. The acceleration-time graph of an object moving in a straight line is shown below. The initial velocity of the object is zero. Find the time when the velocity of object becomes zero again.



(1) 8 s (2) 11.65 s

(3) 12 s (4) 14 s

45. For a particle moving in a straight line, position x (in m) as function of time t (in s) is given by, $x = 25t - 5t^2$. Distance covered by the particle in the 3rd second of its motion is.

(1) zero

(2) 5 m

(3) 10 m

(4) 2.5 m

[Contd...

46. Which of the following species can act as reducing agent?

(1) SO_4^{2-}

(2) CrO_4^{2-}

(3) CO_3^{2-}

(4) NO_2^-

47. The percentage composition of carbon by moles in benzene is:

(1) 25% (2) 50%

(3) 75% (4) 35%

48. For the redox reaction:

$MnO_4^- + C_2O_4^{2-} + H^+ \rightarrow Mn^{2+} + CO_2 + H_2O$

The **correct** coefficients of the reactants for the balanced reaction are:

	MnO_4^-	$C_2O_4^{2-}$	H^+
(1)	5	2	8
(2)	2	5	16
(3)	5	1	8
(4)	2	4	13

49. How many concentration terms among the following are temperature dependent?

Molarity, Molality, Mole fraction, density, % by mass

(1) 2

(2) 3

(3) 4

(4) 5

50. Which of the given reactions is **not** an example of disproportionation reaction?

(1) $2NO_2 + 2NaOH \rightarrow NaNO_2 + NaNO_3 + H_2O$

(2) $2H_2O_2 \rightarrow 2H_2O + O_2$

(3) $Cl_2 + 2NaOH \rightarrow NaCl + NaClO + H_2O$

(4) $I_2 + H_2O_2 + 2NaOH \rightarrow 2NaI + 2H_2O + O_2$

51. Match **List-I** with **List-II**.

List-I		List-II	
(A)	1g H_2(g)	(I)	Weighs 6.4 g
(B)	0.2 mol O_2(g)	(II)	Weighs 28 g
(C)	1 g molecules of N_2(g)	(III)	Contain 13 atoms
(D)	52 u of He	(IV)	Occupies 11.2 L at STP

Choose the **correct** answer from the options given below:

(1) A-III; B-II; C-IV; D-I

(2) A-II; B-I; C-IV; D-III

(3) A-IV; B-I; C-II; D-III

(4) A-IV; B-II; C-I; D-III

52. Which among the following is **not** a mixed oxide?

(1) C_3O_2 (2) Fe_3O_4

(3) Mn_3O_4 (4) Pb_3O_4

53. **Incorrect** statement among the following is:

(1) the SI unit of amount of substance is mole

(2) formula mass is used for NaCl instead of molecular mass.

(3) Dalton's theory could explain the laws of gaseous volumes.

(4) one atomic mass unit is the mass exactly equal to one-twelfth of the mass of one carbon-12 atom.

54. Layer test is used for the identification of:

(1) Cl^- and Br^- ions only

(2) Br^- and I^- ions only

(3) F^- and Cl^- ions only

(4) Cl^-, Br^- and I^- ions

55. Volume of O_2(g) required at STP for complete combustion of 8 g of methane is:

(1) 5.6 L (2) 22.4 L

(3) 11.2 L (4) 2.8 L

56. Given below are two statements:

Statement I: All decomposition reactions are redox reactions.

Statement II: Iodine gives an intense red colour with starch.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

57. Molality of an aqueous solution of urea (NH_2CONH_2) containing 10% urea by mass is:

(1) 2.35 m (2) 1.85 m

(3) 0.65 m (4) 3.15 m

[Contd...

58. Oxidation state of phosphorus in $Ca(H_2PO_2)_2$ is:
(1) +1
(2) +2
(3) +3
(4) $+\frac{1}{2}$

59. Law of multiple proportions is valid for which of following pairs?
(1) N_2O and N_2O_5
(2) N_2 and N_2O
(3) N_2 and O_2
(4) O_2 and O_3

60. Consider the following for the given reaction:
$2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$
A. Redox reaction
B. Displacement reaction
C. Disproportionation reaction
D. Decomposition reaction
The **correct** options for the given reaction are:
(1) A and C only
(2) B, C and D only
(3) A and D only
(4) A, B and C only.

61. Number of significant figures in 2.01×10^2 is:
(1) 3 (2) 5
(3) 2 (4) 4

62. Oxidation state of nitrogen in NH_3, N_3H and N_2H_4 increases in the order:
(1) $N_3H < N_2H_4 < NH_3$
(2) $NH_3 < N_3H < N_2H_4$
(3) $NH_3 < N_2H_4 < N_3H$
(4) $N_2H_4 < N_3H < NH_3$

63. The molecular formula of a compound (molar mass = 99 g) containing 25% carbon, 71% chlorine and 4% hydrogen by mass is:
(1) CH_2Cl_2
(2) $C_2H_4Cl_2$
(3) CH_3Cl
(4) $C_2H_2Cl_2$

64. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Among halogens, F_2 is most powerful oxidising agent.
Reason R: Reduction potential of fluorine is maximum.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

65. 0°C is equivalent to:
(1) 23 °F
(2) 32 °F
(3) 16 °F
(4) 48 °F

66. Match **List-I** with **List-II**.

List-I (Species)		**List-II (Oxidation state of oxygen)**	
(A)	Peroxide	(I)	–2
(B)	Ozonide	(II)	$-\frac{1}{2}$
(C)	Oxide	(III)	$-\frac{1}{3}$
(D)	Superoxide	(IV)	–1

Choose the **correct** answer from the options given below:
(1) A-III; B-II; C-IV; D-I
(2) A-IV; B-III; C-I; D-II
(3) A-III; B-I; C-II; D-IV
(4) A-IV; B-II; C-I; D-III

67. Number of valence electrons in 0.7 g of N^{3-} ion is:
(1) 0.4 N_A (2) 0.25 N_A
(3) 0.2 N_A (4) 0.65 N_A

68. Given below are two statements:
Statement I: The standard electrode potential of hydrogen electrode is 0.00 volts.
Statement II: The oxidation state of hydrogen is +1 in all of its compounds.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

[Contd...

69. Mass percentage of carbon in CO_2 is:
(1) 14.83%
(2) 32.35%
(3) 21.18%
(4) 27.27 %

70. The electron releasing tendency of the given metals is:
(1) Zn > Cu > Ag
(2) Cu > Zn > Ag
(3) Ag > Cu > Zn
(4) Ag > Zn > Cu

71. Mass of one molecule of SO_2 gas at STP is:
(1) 64 g
(2) $\frac{64}{N_A}$ g
(3) $\frac{N_A}{64}$ g
(4) 64 N_A g

72. Using stock notation, the **correct** representation of $HAuCl_4$ is:
(1) H(III)$AuCl_4$
(2) HAu(III)Cl_4
(3) $HAuCl_4$(III)
(4) (III)$HAuCl_4$

73. Match **List-I** with **List-II**.

List-I (Multiple)		List-II (Prefix)	
(A)	10^{-2}	(I)	Micro
(B)	10^{-6}	(II)	Pico
(C)	10^{-12}	(III)	Centi
(D)	10^{-9}	(IV)	Nano

Choose the **correct** answer from the options given below:
(1) A-III; B-II; C-IV; D-I
(2) A-IV; B-III; C-I; D-II
(3) A-II; B-IV; C-I; D-III
(4) A-III; B-I; C-II; D-IV

74. Number of mole of $K_2Cr_2O_7$ required to oxidise one mol of I^- into I_2 in acidic medium is:
(1) $\frac{1}{3}$
(2) 3
(3) 6
(4) $\frac{1}{6}$

75. If 1.8×10^{-3} mol of H_2 is left after removing 3.011×10^{21} molecules of H_2 from its sample, the initial mass of H_2 sample is:
(1) 13.6 mg
(2) 6.8 mg
(3) 3.4 mg
(4) 1.52 mg

76. Match **List-I** with **List-II**.

List-I (Reaction)		List-II (Example)	
(A)	Metal displacement redox reaction	(I)	$2Na + 2H_2O \rightarrow 2NaOH + H_2$
(B)	Combination reaction	(II)	$NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
(C)	Non-redox reaction	(III)	$CuSO_4 + Zn \rightarrow Cu + ZnSO_4$
(D)	Non-metal displacement redox reaction	(IV)	$C + O_2 \rightarrow CO_2$

Choose the **correct** answer from the options given below:
(1) A-III; B-IV; C-II; D-I
(2) A-IV; B-III; C-I; D-II
(3) A-II; B-IV; C-I; D-III
(4) A-III; B-I; C-II; D-IV

77. The **correct** postulates of the Dalton's atomic theory are;
(A) Compounds are formed when atoms of different element combine in a fixed ratio.
(B) Chemical reactions involve reorganisation of atoms.
(C) All atoms of given element have identical properties.
(D) Matter consists of divisible atoms.
(E) Atoms can be created or destroyed in a chemical reaction.
Choose the **correct** answer from the options given below:
(1) A, B, C only
(2) A, B, D only
(3) B, C, D only
(4) C, D, E only

78. The oxidation state of Br atoms in Br_3O_8 are:
(1) +4 and +6
(2) +5 and +6
(3) +6 and +7
(4) +4 and +5

79. The molarity of HCl solution containing 36.5% HCl by mass and density 1.4 g mL^{-1} is:
(1) 7 M (2) 10.5 M
(3) 14 M (4) 17.5 M

80. Which of the following oxidation state carbon can **not** exhibit?
(1) +4 (2) –5
(3) Zero (4) –1

81. Number of atoms in the following samples of substance is largest in:
(1) 4 g hydrogen gas
(2) 32 g ozone
(3) 71 g chlorine gas
(4) 2.3 g sodium

82. Oxidation can be defined as:
(A) loss of electron(s) by any species
(B) addition of electronegative element to a substance
(C) removal of hydrogen from a substance
Choose the **correct** option:
(1) A and B only.
(2) A and C only
(3) B and C only
(4) A, B and C

83. Consider the reaction:
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
In which of the following combinations, H_2 act as limiting reagent:
(1) 28 g N_2 + 12 g H_2
(2) 14 g N_2 + 2 g H_2
(3) 7 g N_2 + 4 g H_2
(4) 28 g N_2 + 8 g H_2

84. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Fluorine does not show a disproportionation tendency.
Reason R: Fluorine is a non-metal.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

85. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Carbon exist as mixture of three isotopes ^{12}C, ^{13}C and ^{14}C.
Reason R: Average atomic mass of carbon is 12.011 u.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

86. Reductant and oxidant in the reaction, $2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$, respectively are:
(1) Sulphur of Cu_2S and Cu(II)
(2) Copper of Cu_2S and Cu(I)
(3) Sulphur of Cu_2S and Cu(I)
(4) Copper of Cu_2S and Cu(II)

87. On diluting 100 mL of 0.2M NaOH solution by 900 mL of water, the mole of NaOH in resulting diluted solution will be:
(1) 0.18 mol
(2) 0.0009 mol
(3) 0.002 mol
(4) 0.02 mol

88. The oxidation number of sulphur in S_2F_2 is:
(1) Zero
(2) +1
(3) +2
(4) –1

89. The amount of oxygen in 0.2 mol of water is:
(1) 1.6 g
(2) 3.2 g
(3) 0.8 g
(4) 5.4 g

90. In CrO_5, Oxidation state of Cr is:
(1) +10
(2) +8
(3) +6
(4) +5

91. Read the following statements regarding the nucleus.

A. The outer nuclear membrane is usually continuous with the endoplasmic reticulum.
B. The movement of RNA and protein molecules is bidirectional through nuclear pores.
C. The nucleolus is a membrane-bound structure.
D. Present only in eukaryotic cells.
E. Larger and more numerous nucleoli are found in cells actively synthesising proteins.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) A, C and E are incorrect
(2) A, B, D and E are correct
(3) B, C and E are correct
(4) C, D and E are incorrect

92. Which of the following is not present in the main arena of various cellular activities of most plant cells?

(1) Plastids (2) Vacuole
(3) Centrioles (4) Mitochondria

93. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Chromatophores	(I)	Contain fat soluble carotenoid pigments in plant cells
(B)	Chromoplasts	(II)	Held together by centromere in a chromosome
(C)	Chromatids	(III)	A loose and indistinct network of nucleoprotein fibres in interphase nucleus
(D)	Chromatin	(IV)	Contain pigments in cyanobacteria

Choose the **correct** answer from the options given below.

(1) A-III, B-I, C-II, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

94. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The cell membrane's fluidity is crucial for functions like cell growth and secretion.

Reason (R): The quasi-fluid nature of proteins enables the lateral movement of lipids within the plasma membrane.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

95. Typically, the number of centromere(s) in a metacentric and acrocentric chromosome is;

(1) one.
(2) two.
(3) three.
(4) four.

96. Given below are two statements:

Statement I: Matthias Schleiden, a German zoologist, observed that all plants are composed of different kinds of cells.

Statement II: Theodore Schwann, a British botanist, studied animal cells and reported the presence of a cell wall as a unique character.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

97. The centriole forms the basal body of a flagellum. In contrast to the flagellum structure with 9+2 microtubular arrangement, the centriole is composed of:

(1) 9 doublets of radially arranged peripheral microtubules.
(2) 27 microtubules.
(3) a core called the axoneme.
(4) 2 centrally located microtubules.

98. Select the **incorrectly** matched pair.

(1) Mesosome – DNA replication
(2) Pili – Motility
(3) Fimbriae – Attachment to host tissues
(4) Polysome – Protein synthesis

99. Shown below is a cell organelle. Choose the **correct** statement about it.



(1) It is found in plant cell but not in bacterial cell.
(2) It is composed of ribonucleic acid and proteins and is surrounded by a single membrane.
(3) In an animal cell, its function is carbohydrate synthesis.
(4) Its two subunits are similar in the mitochondria and chloroplasts.

100. Read the following statements about the cell envelope of a prokaryotic cell.
A. It consists of a loosely bound three-layered structure.
B. The layers are glycocalyx, cell wall and mesosome.
C. The three layers act together as a single protective unit.
D. The cell wall prevents the bacterium from bursting or collapsing.
E. Glycocalyx is always a thick and tough structure.
Choose the option with the **incorrect** statements.
(1) A, C and D (2) A, B and E
(3) B, C and D (4) A, D and E

101. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): RER is frequently observed in cells actively involved in protein synthesis and secretion.
Reason (R): The SER is the major site for synthesis of lipid.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

102. While observing a cell, a scientist saw two chromosomes viz. A and B in it. In 'A', satellite was observed which was not seen in 'B'. Choose the incorrect statement.
(1) Both 'A' and 'B' have primary constriction.
(2) In 'B' secondary constriction is present.
(3) In both 'A' and 'B', the centromere has disc-shaped structures on the sides.
(4) The cell is likely to be in a dividing state.

103. Identify the **incorrect** statement about 'Reductionist Biology' approach to study and understand living organisms.
(1) It is a physico-chemical approach.
(2) Cell-free systems cannot be used in this approach of study.
(3) It can explain the abnormal processes that occur during any diseased condition.
(4) This approach enables us to describe the various processes of a cell in molecular terms.

104. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Golgi apparatus	(I)	Divides the intracellular space into luminal and extra luminal compartments
(B)	Lysosomes	(II)	Contains water, sap, excretory product and other waste materials of a cell
(C)	Vacuole	(III)	Formation of glycoproteins
(D)	Endoplasmic reticulum	(IV)	Possess almost all types of hydrolytic enzymes

Choose the **most appropriate** answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

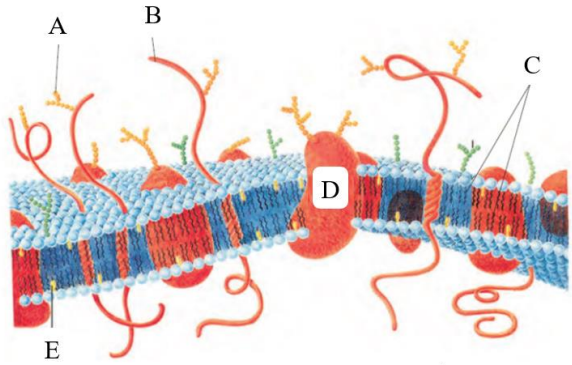
105. Given below are two statements:
Statement I: The average diameter of a mitochondrion is 0.5 nm.
Statement II: The inner membrane of mitochondria forms a number of infoldings called the cristae towards the matrix.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

106. Identify the **correct** statement.
(1) The organisation of the centrosome is described as a 'cartwheel'.
(2) Each centrosome is surrounded by an amorphous pericentriolar material.
(3) Each centrosome consists of two centrioles.
(4) The centrioles in a centrosome are cylindrical and arranged parallel to each other.

[Contd...

107. Which of the given statements are true for chloroplasts?
A. They are found in euglenoids.
B. They are double membrane bound, with the inner membrane being more permeable.
C. The stroma contains enzymes for carbohydrate and protein synthesis.
D. The ribosomes of the chloroplasts are 80S type.
E. Chlorophyll pigments are located in the stroma.
Choose the **correct** option.
(1) A and C only (2) A, B and C
(3) C and D only (4) B and E only

108. Given below is the representation of plasma membrane structure. Choose the correct statement about it.



(1) B is comparatively easier to extract than D.
(2) In C, the non-polar heads are exposed to the aqueous environment.
(3) E is a polysachharide.
(4) A is synthesised inside mitochondria.

109. Choose the **correctly** matched pair from the following.
(1) Rudolf Virchow - Discovered the nucleus
(2) Camillo Golgi - Discovered ribosomes
(3) Singer and Nicolson - Proposed fluid mosaic model of plasma membrane
(4) Robert Brown - Formulated cell theory

110. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Lysosomes are formed by the process of packaging in the golgi apparatus.
Reason (R): Lysosomes are single membrane bound organelles.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

111. Choose the **odd one out** for mesosomes.
(1) Formed by infoldings of plasma membrane.
(2) Help in cell wall formation in prokaryotic cells.
(3) Are extra genomic DNA molecules.
(4) Are in the form of vesicles, tubules and lamellae.

112. Identify the set of **correct** statements.
A. The endomembrane system includes ER, golgi complex, lysosomes and vacuoles.
B. The functions of mitochondria and chloroplasts are coordinated with the endomembrane system.
C. RER bears ribosomes on its outer surface.
D. Proteins synthesized by ribosomes on the ER are modified in the cisternae of the Golgi apparatus.
E. Peroxisome is a part of endomembrane system.
Choose the **correct** option.
(1) A, B, C (2) A, C, D
(3) B, C, D (4) C, D, E

113. Match **List-I** with **List-II**.

List-I		List-II	
(A)	DNA	(I)	Confers antibiotic resistance in bacteria
(B)	Nucleus	(II)	Present in nucleus, mitochondria and chloroplast
(C)	Plasmid	(III)	Not present in mature sieve tube cell
(D)	Microbodies	(IV)	Single-membrane bound vesicles having enzymes

Choose the **most appropriate** answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

114. The cell wall of a newly discovered microorganism is made of cellulose, galactans, mannans and minerals like calcium carbonate. Most likely, this organism;
(1) is a prokaryote.
(2) possesses a well-defined nucleus.
(3) is unable to divide.
(4) lacks cell membrane.

115. Given below are two statements:
Statement I: Na^+/K^+ exchange pump transports these ions across the plasma membrane against the concentration gradient.
Statement II: Na^+/K^+ exchange pump does not require energy for transport of these ions across the plasma membrane.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

[Contd...

116. Choose the option that includes all eukaryotic organisms only.
(1) Protists, plants and animals.
(2) Cyanobacteria, fungi and algae.
(3) Plants, animals and mycoplasma.
(4) Protists, animals and bacteria.

117. Choose the **incorrect** statement.
(1) ER helps in the synthesis of glycogen.
(2) ER helps in the synthesis of lipoproteins.
(3) ER helps in the transport of substances.
(4) ER helps in the osmoregulation and excretion in *Amoeba*.

118. Read the following statements.
A. Mycoplasmas are about 3-5 µm in length.
B. Bacteria can have a rod-like shape called bacillus.
C. The largest isolated single cell is the egg of an ostrich.
D. Human red blood cells are about 7.0 µm in diameter.
E. Nerve cells are the smallest cells in multicellular organisms.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) A, B, C are incorrect
(2) B, C, D are correct
(3) C, D, E are incorrect
(4) A, C, E are correct

119. A typical red blood cell;
A. is round and oval in shape.
B. possesses two nuclei at maturity.
C. has 52% proteins in its plasma membrane.
D. has 40% lipids in its plasma membrane.
E. lacks cell wall.
The **incorrect** statements are;
(1) C and D (2) D and E
(3) A and C (4) A and B

120. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The two membranes of the mitochondrion have their own specific enzymes.
Reason (R): Mitochondria are called the powerhouses of the cell.
In the light of the above statements, choose the most appropriate answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

121. The cell is considered as the structural and fundamental unit of life in all living organisms because;
(1) unicellular organisms are not capable of independent existence.
(2) unicellular organisms are not capable of performing the essential functions of life.
(3) anything less than a complete structure of a cell does not ensure independent living.
(4) division of labour is seen in multicellular organisms.

122. Arrange the following cells/structures in **correct** order of their size.
Virus, Typical bacteria, Typical eukaryotic cell, Golgi cisternae
(1) Virus > Typical bacteria > Typical eukaryotic cell > Golgi cisternae
(2) Virus < Typical bacteria < Typical eukaryotic cell < Golgi cisternae
(3) Typical eukaryotic cell > Typical bacteria > Golgi cisternae > Virus
(4) Typical eukaryotic cell < Typical bacteria < Golgi cisternae < Virus

123. How many of the following possess cell wall?
Onion cell, Fungi, White blood cell, Mycoplasma
(1) One
(2) Two
(3) Three
(4) Four

124. Match **List-I** with **List-II**.

List-I (Type of chromosome)		**List-II (Characteristic)**	
(A)	Telocentric	(I)	One shorter arm and one longer arm
(B)	Sub-metacentric	(II)	Two equal arms
(C)	Metacentric	(III)	Terminal centromere
(D)	Acrocentric	(IV)	One extremely short and one very long arm

Choose the **most appropriate** answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

125. The chromatin;
(1) is stained by acidic dyes.
(2) was given the name by Flemming to nuclear material.
(3) is present in PPLO.
(4) lacks RNA.

[Contd...

126. Given below are two statements:
Statement I: A single human cell has approximately two metres long thread of DNA.
Statement II: The two metres long DNA is distributed among twenty-three chromosomes in a single human cell.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

127. Choose the **correct** sequence of the layers in bacteria cell envelope from inside to outside.
(1) Glycocalyx – Cell wall – Cell membrane
(2) Cell wall – Glycocalyx – Cell membrane
(3) Cell wall – Cell membrane – Glycocalyx
(4) Cell membrane – Cell wall – Glycocalyx

128. Choose the **incorrect** statement.
(1) Bacteria can be classified into two groups on the basis of the differences in the cell envelopes and the manner in which they respond to the staining procedure developed by Gram.
(2) The four basic shapes of bacteria are bacillus, coccus, vibrio and spirillum.
(3) Gram positive bacteria do not take up Gram stain.
(4) The organisation of the prokaryotic cell is fundamentally similar even though prokaryotes exhibit a wide variety of shapes and functions.

129. Which of the following statements about prokaryotic cells are **incorrect**?
A. They are generally smaller and multiply more rapidly than eukaryotic cells.
B. The genetic material is basically naked.
C. The genome is made up of two chromosomes/circular DNA.
D. Plasmid DNA is used to monitor bacterial transformation with self DNA.
E. Their plasma membrane is structurally different from that of eukaryotes.
Choose the **correct** option.
(1) A and B only (2) C, D and E
(3) A, C and D (4) A, D and E

130. In a plant cell, the;
(1) cell wall is never traversed by plasmodesmata.
(2) primary cell wall is unable to grow.
(3) secondary cell wall is formed on the outer side of primary wall.
(4) middle lamella is mainly made up of calcium pectate.

131. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Thylakoids	(I)	The matrix of nucleus
(B)	Matrix	(II)	Flattened membranous sacs in stroma of chloroplast
(C)	Nucleoplasm	(III)	Dense homogenous substance in the inner compartment of mitochondria
(D)	Cytoplasm	(IV)	Semi-fluid matrix that fills the cell

Choose the **most appropriate** answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-IV, C-II, D-I

132. Identify the **incorrect** statement.
(1) The grana, in the plastid, is the site of light reactions and the stroma of dark reactions.
(2) Microvilli are finger-like projections in the plant cells.
(3) Leucoplasts are the colourless plastids of varied shapes and sizes.
(4) Amyloplasts store starch.

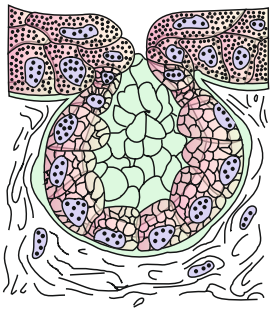
133. Consider the following statements about cell inclusions.
A. They store reserve materials in prokaryotic cells.
B. They are bound by a single membrane.
C. They lie free in the cytoplasm.
D. Phosphate granules are an example of inclusion bodies.
E. Gas vacuoles are not found in any bacteria.
The **correct** statements are;
(1) A, B and D (2) B, D and E
(3) A, C and D (4) A, C and E

134. Which of the following statements **correctly** describes the *'cis'* and *'trans'* faces of the Golgi apparatus?
(1) The *cis* face is the maturing face, and the *trans* face is the forming face.
(2) Both faces are identical in structure and function but are located at opposite ends.
(3) The *cis* face is the convex face, and the *trans* face is the concave face.
(4) Vesicles from the ER fuse with the *trans* face and exit from the *cis* face.

135. How do eukaryotic cells maintain their shape and carry out motility?
(1) Solely through the rigidity of the plasma membrane.
(2) Through an elaborate network of filamentous proteinaceous structures called the cytoskeleton.
(3) By the pressure exerted by the large central vacuole.
(4) Through the action of cilia and flagella only.

[Contd...

136. Select the **correct** option for the diagram of gland.



	Gland	Location	Function
(1)	Unicellular	GIT	Secrete mucus
(2)	Multicellular	Mouth	Secrete saliva
(3)	Unicellular	Buccal cavity	Secrete saliva
(4)	Multicellular	Skin	Secrete mucus

137. Read the statements about cockroach anatomy and identify the **incorrect** one.

(1) Mosaic vision is common at night, hence called nocturnal vision.

(2) In cockroach, the brain supplies nerves to the antennae and compound eyes.

(3) Fat body, nephrocytes and uricose glands also aid in excretion.

(4) Male cockroach bears one pair of collateral glands functioning as accessory reproductive glands.

138. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Chitinous exoskeleton	(I)	Segmental arrangement in both sexes, shows 10 divisions
(B)	Thorax	(II)	Carries two pairs of wings
(C)	Abdomen	(III)	Provides mechanical support and protection
(D)	Jointed legs	(IV)	Enables locomotion with flexibility

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-I, D-IV

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-II, C-IV, D-I

139. Read the following statements and find out the **incorrect** statement about frog.

(1) The blood cells are RBC, WBC and platelets.

(2) The lymph is different from blood as it lacks few proteins and RBCs.

(3) Kidneys are bean shaped structures, of which one is situated a little anteriorly in the body cavity and they are present on both sides of vertebral column.

(4) In male frog, ureters act as urinogenital duct, which opens into cloaca.

140. How can a male frog be distinguished from a female frog?

(1) Presence of vocal sacs and a copulatory (nuptial) pad on the first digit of the forelimb.

(2) Absence of neck and tail.

(3) Hind limb ending in five digits.

(4) Bulged eyes covered by a nictitating membrane.

141. Why do cartilage tissues heal slowly after injury?

(1) Cartilage cells cannot reproduce.

(2) They lack direct blood supplies.

(3) The intercellular material is missing.

(4) Cartilage cells are surrounded by fluids.

142. Given below are two statements:

Statement I: On land, adult frogs use skin, lungs, and the bucco-pharyngeal cavity for breathing.

Statement II: During aestivation and hibernation, gaseous exchange occurs through the skin in frog.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

143. Consider these statements about stratified epithelium.

A. It has a limited role in secretion and absorption.

B. It mainly forms a diffusion boundary.

C. It lines blood vessels.

D. Only the lowermost cell layer contacts the basement membrane.

How many of the statements is/are **incorrect?**

(1) One (2) Two

(3) Three (4) Four

[Contd...

144. Read the following statements carefully w.r.t. the main circulating fluid in humans.

A. Structural proteins are present in plasma.
B. Its cell secretes modified polysaccharide that act as intercellular substances.
C. It helps exchange nutrients between tissues and blood therefore, it is also known as lymph.
D. Most of the cells are enucleated.

Choose the option with **incorrect** statement(s) only.

(1) A only
(2) A and B only
(3) A, B and C only
(4) A, B, C and D

145. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Cockroaches are known as 'friends of farmers' due to their role in pest control.

Reason R: As omnivores, cockroaches are pests that destroy and contaminate stored grains and crops, and they can also act as vectors for various diseases.

In the light of the above statements, choose the **correct** answer from the options given below.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both A and R are true, but R is not the correct explanation of A.
(3) A is true, but R is false.
(4) A is false, but R is true.

146. In a simple epithelium, the cells are:

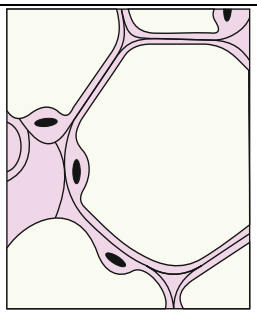
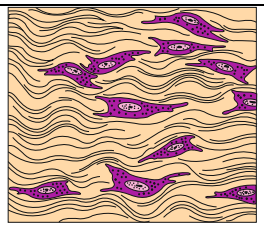
(1) Incapable of division to replace old and damaged cells.
(2) Hardened to provide structural support to organs.
(3) Loosely connected, forming an irregular layer.
(4) Cemented closely to one another, forming a single layer.

147. Consider the following four statements (A-D) related to the common frog *Rana tigrina,* and select the **correct** option stating which ones are true (T) and which ones are false (F).

A. On dry land it would die due to lack of O_2 if its mouth and nostrils are forcibly kept closed for a few days.
B. It has four chambered heart.
C. On dry land it turns uricotelic from ureotelic.
D. Its tadpole lives in pond water.

	A	B	C	D
(1)	F	F	T	T
(2)	T	F	F	T
(3)	F	T	T	F
(4)	T	T	F	F

148. Match **List-I** with **List-II.**

List-I		List-II	
(A)		(I)	Connect skin with muscle
(B)		(II)	Present in skin
(C)		(III)	Arrangement of fibres in ligament
(D)		(IV)	Triglyceride storage

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-I D-II
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-I, C-II, D-III

149. The ciliated epithelial cells are required to move particles or mucus in a specific direction. In humans, these cells are mainly present in;

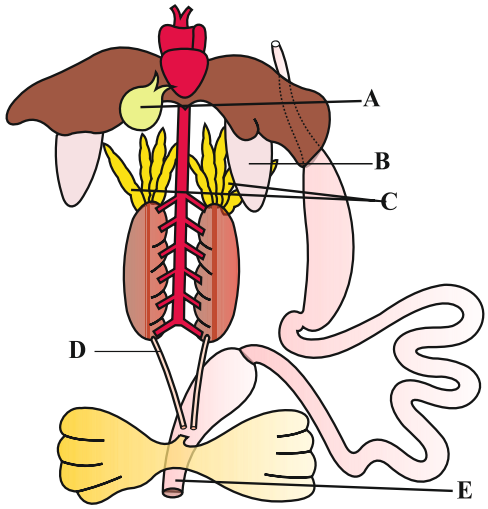
(1) Fallopian tubes and Pancreatic duct
(2) Eustachian tube and Salivary duct
(3) Bronchioles and Fallopian tubes
(4) Bile duct and Bronchioles

150. With respect to respiratory system of cockroach, one amongst the following statements is **wrong.**

(1) First and second thoracic spiracles always remain open.
(2) Tergo-sternal muscles aid in respiration of cockroaches.
(3) Tracheal tubes are supported by chitinous rings.
(4) Respiratory pigment is not involved in transfer of gases to tissues.

[Contd...

151. The figure given here shows diagrammatic representation of internal organs of frog. Identify A to E and select the **correct** option.



	A	B	C	D	E
(1)	Gall bladder	Lungs	Ovary	Renal artery	Rectum
(2)	Liver	Pancreas	Fat bodies	Testis	Gallbladder
(3)	Pancreas	Lung	Testis	Kidney	Liver
(4)	Gall bladder	Lung	Fat bodies	Ureter	Cloaca

152. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Pseudostratified epithelium gives a false appearance of being stratified.
Reason R: Pseudostratified epithelium is composed of more than one layer of cells to provide protection against stress.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

153. Find the **incorrect** statement.
(1) Cockroaches belong to the class Insecta of phylum Arthropoda.
(2) In cockroaches, body is divided into three distinct regions: head, thorax and abdomen.
(3) In *P. americana*, triangular head lies anteriorly at acute angles to the longitudinal body axis.
(4) In frog, bile emulsifies fat and pancreatic juices, digest carbohydrates and proteins.

154. Which of the following statements **correctly** describes the formation and function of an organ system in complex multicellular animals?
(1) An organ system is formed when a single type of tissue performs all the necessary functions for the body's survival, exhibiting a high degree of cellular specialisation.
(2) Organ systems arise from the coordinated physical and/or chemical interactions of two or more organs that share a common function, contributing to the overall division of labor.
(3) The fundamental basis of an organ system is a group of similar cells and their intercellular substances, working together to achieve a specific function.
(4) Complex animals are characterised by having billions of cells, each capable of performing all life functions independently within an organ system.

155. Which of the following type of tissue is being described by the given statements?
- They are named because of their special function of linking and supporting other tissues/organs of the body.
- They include cartilage, bone, adipose and blood.
- They provide strength, elasticity and flexibility to the tissue.
- Cells of these tissues secrete fibres of structural proteins called collagen or elastin.

(1) Epithelial tissue
(2) Connective tissue
(3) Muscle tissue
(4) Neural tissue

156. A patient in a coma, who is otherwise physiologically stable (e.g., heart is beating, kidneys are filtering), is unable to respond to external stimuli like sound or touch. This clinical scenario provides evidence that:
(1) The patient's muscle tissues have lost their ability to contract.
(2) The patient's connective tissues are no longer supporting the organs.
(3) The patient's epithelial tissues have failed to perform their secretory functions.
(4) The integrative and communication functions of the patient's neural tissue are impaired, preventing a coordinated response.

[Contd...

157. Consider the following statements:
Statement I: Eyes and internal ears are well-organised structures in frog and the rest sensory organs of touch and taste are cellular aggregations around muscular endings.
Statement II: Division of labour among cells, tissues and organs increases metabolic redundancy and slows survival responses.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I and Statement II both are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Statement I and Statement II both are incorrect.

158. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Gall bladder in frog	(I)	Mammalian bone
(B)	Dendrites	(II)	Enclosed in a sheath of connective tissue
(C)	Bundle of muscle fibres	(III)	Temporary storage reservoir for bile produced by the liver
(D)	Haversian canal	(IV)	Nerve cells

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-IV, D-II
(2) A-III, B-I, C-IV, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-IV, C-II, D-I

159. Carefully read the following statements and identify how many of the following statements are **correct**?
A. The secretion of tears, milk, sweat and oil are functions of epithelial tissue.
B. Exposed surface of the human body is covered by the stratified squamous epithelium.
C. The digestive system of a cockroach includes two pair of salivary glands.
D. The primary function of the salivary reservoir is to store the saliva that is produced by the salivary glands.
E. The antennae of a cockroach are long, thread-like, and multisegmented, providing a high degree of sensory input from the environment. This type of antenna is correctly termed as moniliform antenna.
(1) Only three statements are correct
(2) Only four statements are correct
(3) All five statements are correct
(4) Only two statements are correct

160. Match **List-I** (structures/functions) with **List-II** (corresponding features) and select the **correct** combination.

List-I		List-II	
(A)	Prevents leakage of body fluids between adjacent cells	(I)	Gap junctions
(B)	Provides cytoplasmic continuity between neighbouring cells	(II)	Ligaments
(C)	Acts as anchoring points that bind cells together	(III)	Tight junctions
(D)	Connects one bone to another	(IV)	Adhering junctions

(1) A-III, B-II, C-I, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-III, B-I, C-II, D-IV
(4) A-III, B-I, C-IV, D-II

161. Given below are two statements:
Statement I: Transitional epithelium is a kind of simple epithelium.
Statement II: The bright yellow, red, and green colors reported in tropical cockroach species are an example of sexual dimorphism, being found exclusively on the male's tegmina (forewings) to attract mates.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

162. The mechanism of pulmonary respiration in a frog involves creating positive pressure in the buccal cavity to force air into the lungs. If a perforation is made in the floor of a frog's buccal cavity, the frog can survive for a considerable period. This survival is primarily attributed to:
(1) The enhancement of buccal respiration due to increased airflow into the oral cavity.
(2) The continued, unimpeded function of the pulmonary respiratory cycle.
(3) The ability of its skin (cutaneous respiration) to perform gaseous exchange independently of the buccal pump mechanism.
(4) The activation of anaerobic metabolism in its muscles to compensate for the temporary halt in breathing.

[Contd...

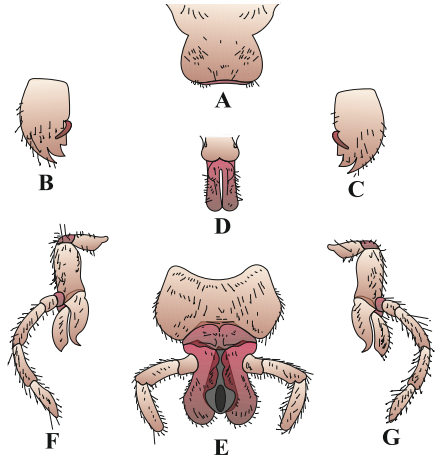
163. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Mast cells	(I)	Tip of nose
(B)	Hyaline cartilage	(II)	Areolar connective tissue
(C)	Crop	(III)	Cuboidal epithelium
(D)	Tubular parts of nephron	(IV)	Temporary storage of food

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-II, B-I, C-IV, D-III

164. Following is the diagram of mouth parts of cockroach.



Which of the following labelled structures will help in grinding and chewing of food?

(1) A and D (2) B and C
(3) A and C (4) D and E

165. Consider the following statements.

A. Each muscle is made of many long, cylindrical fibres arranged in parallel arrays.
B. These fibres are composed of numerous fine fibrils, called neurofibrils.
C. Muscle fibres contract (shorten) in response to stimulation, then relax (lengthen) and return to their contracted state in a coordinated fashion.
D. Their action moves the body to adjust to the changes in the environment and to maintain the positions of the various parts of the body.
E. In general, muscles play an active role in few of the movements of the body.

Which of the above statements are **correct**?

(1) A, B and E only
(2) A, C and D only
(3) A and D only
(4) B and E only

166. Match **List-I** with **List-II** w.r.t. cockroach.

List-I		List-II	
(A)	Mouth parts of Cockroach	(I)	Anal cerci
(B)	Segments of walking leg	(II)	Malpighian tubule
(C)	Sound receptor structure	(III)	Mandibles, maxillae
(D)	Excretory unit	(IV)	Coxa, trochanter

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-III, D-IV
(2) A-III, B-IV, C-II, D-I
(3) A-III, B-IV, C-I, D-II
(4) A-IV, B-III, C-I, D-II

167. Read the following statement.

A. Nervous system of cockroach consists of a series of segmentally fused ganglia located on dorsal side of body.
B. In cockroach, sensory structures are absent on maxillary and labial palps.
C. Supra oesophageal ganglion function as brain in cockroach
D. Mosaic vision in cockroach has less sensitivity but resolution is high

Select the **correct** option stating which statement is true and which one are false.

	A	B	C	D
(1)	T	T	T	F
(2)	T	F	T	F
(3)	F	F	T	F
(4)	T	T	F	T

168. Identify the **wrong** matching.

(1) Pronotum– Dorsal sclerites of prothorax
(2) Oesophagus– Narrow tubular passage which in turn opens into a sac-like structure(crop)
(3) Ocellus - To form a high-resolution, sharp image.
(4) Gizzard – Bears structure called teeth

169. Considering the structural organization of the stomach, lung, heart, and kidney, which statement is **incorrect**?

(1) All of these organs are composed of epithelial, connective, muscular, and nervous tissue.
(2) Each of these structures is correctly classified as an organ.
(3) While the stomach and heart exhibit all four tissue types, the lung and kidney are primarily composed of only two tissue types.
(4) Each of these organs shows a division of labor among its constituent tissues.

[Contd...

170. Consider the following statements about cockroaches:

A. *Periplaneta americana* is a common species found in India.

B. Decapitation leads to immediate death due to cessation of breathing.

C. The head holds only a small fraction of the nervous system.

D. Sclerites are joined by arthrodial membrane.

How many of the above statements is/are **correct**?

(1) One

(2) Two

(3) Three

(4) Four

171. Given below are two statements:

Statement I: Adipocytes are signet ring shaped in structure and have marginalised or peripheral nucleus.

Statement II: The circulatory system of a cockroach features a dorsal heart composed of 13 funnel-shaped chambers, with its rhythmic pulsation supported by 12 pairs of alary muscles.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

172. Consider the following statements about the Indian bull frog:

A. Saliva plays a crucial role in the chemical digestion of carbohydrates in the buccal cavity.

B. The heart's pacemaker is composed of specialized neural tissue located outside the heart wall.

C. A distinct neck is absent, which contributes to its saltatorial (jumping) locomotion.

D. The primary mode of water intake is through direct ingestion via the mouth.

How many of the above statements are **incorrect**?

(1) One

(2) One pair

(3) Three

(4) Two pair

173. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The mobility of the cockroach head is primarily attributed to the rigidity of its cervical region.

Reason R: The head capsule of *Periplaneta americana* develops from six embryonic segments and is connected to the prothorax through a flexible, membranous neck articulation.

In the light of the above statements, choose the **correct** answer from the options given below:

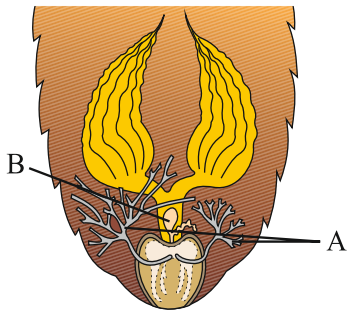
(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false.

(4) A is false but R is true.

174. Following is diagrammatic representation of female reproductive system of cockroach.



Find the **correct** marking and function performed by that part

(1) A - Conglobate gland – Covering of sperm; B - Spermatheca – Storage of sperm

(2) A - Phallomere – Covering around egg; B - Titillator – Opening of spermatophore

(3) A - Vestibule – Place to store fertilized eggs; B - Spermatophore – Storage of sperms

(4) A - Collateral gland – Form the wall of ootheca; B - Spermatheca – Storage of sperms in female

175. Read the following statements regarding wings of cockroach and identify the **incorrect** one.

(1) The hind wings of a cockroach are kept folded beneath the forewings when at rest.

(2) The forewings are opaque and leathery, while the hind wings are transparent and membranous.

(3) The membranous hind wings are the structures that generate lift and are used for flight.

(4) Forewings are known as tegmina and they arise from metathorax.

[Contd...

176. In the reproductive system of a frog, which of the following statements **correctly** describes the path of gametes?

(1) In males, sperms pass from the testes into the vasa efferentia, then directly into the cloaca through the ejaculatory duct.

(2) In females, ova are released from the ovary, travel down the oviduct, and are temporarily stored in the urinary bladder before being expelled.

(3) In males, sperm travel from the testes through the vasa efferentia which enter the kidney, and are then carried by the ureter to the cloaca.

(4) In females, ova are shed into the coelom, collected by the ostium, and then pass through the kidney before entering the cloaca.

177. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Moist surface of pharynx	(I)	Non ciliated columnar epithelium
(B)	Larger ducts of salivary glands	(II)	Ciliated columnar epithelium
(C)	Ventricles of Brain	(III)	Stratified cuboidal epithelium
(D)	Lining of stomach	(IV)	Stratified non keratinised squamous epithelium

Choose the **correct** answer from the options given below:

(1) A-IV, B-III, C-I, D-II

(2) A-IV, B-III, C-II, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-IV, B-II, C-I, D-III

178. Select the **correctly** matched pair.

(1) Small tubules and long tubules in male cockroach system – Secrete substances that nourish sperm and form the spermatophore.

(2) Medulla oblongata in cockroach – Foramen magnum

(3) Skin of frog on ventral side – Olive green with irregular spots.

(4) Walking legs in cockroach – 6 pairs

179. Which of the following statements are **correct**?

A. When a neuron is suitably stimulated, an electrical disturbance is generated which swiftly travels along its plasma membrane.

B. Neuroglia make up more than one-half the volume of neural tissue in our body.

C. Arrival of the disturbance at the neuron's endings, or input zone, triggers events that may cause stimulation or inhibition of adjacent neurons and other cells.

D. Neurons, the unit of neural system are excitable and contractable cells.

(1) A, B and C

(2) B, C and D

(3) A and B only

(4) C and D only

180. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The oxygen-carrying capacity of a given volume of mammalian blood is significantly higher than that of the same volume of frog blood.

Reason R: This is because mature mammalian erythrocytes are enucleated, which allows for maximum intracellular volume to be packed with haemoglobin, whereas the presence of a nucleus in frog erythrocytes limits their haemoglobin content.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true, and R is the correct explanation of A.

(2) Both A and R are true, but R is not the correct explanation of A.

(3) A is true, but R is false.

(4) A is false, but R is true.

■■■

SPACE FOR ROUGH WORK



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if